Simulating Robotic Ultrasound via Contact-Aware RL

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Introduction

Project Aim:

To identify which simulation and reward components are required to achieve reliable zero-shot robotic ultrasound scanning from simulation in (Unity) to real tissue.

Reinforcement Learning:

- Trial-and-error learning where the agent takes actions to maximize a cumulative reward. [1]
- The robotic probe learns to autonomously scan for foreign bodies, maintain contact, and minimize tissue deformation.

Zero-Shot Transfer:

- RL policy trained entirely in simulation and deployed on real life robots and phantom tissues without real-world tuning.
- Evaluates how well simulation-trained policies generalize to real-world ultrasound scanning.

Deformation and Contact-Aware Simulation:

- Simulates soft-tissue deformation and contact forces for realistic probe–tissue interaction. [2]

Methods

Configuration File:

```
behaviors:
  CurvilinearProbe:
    trainer_type: sac
    hyperparameters:
      learning_rate: 0.001
      batch_size: 512
      buffer_size: 100000
      buffer_init_steps: 10000
      tau: 0.005
      steps_per_update: 20
      init_entcoef: 0.3
      learning_rate_schedule: constant
    network_settings:
      vis_encode_type: resnet
      normalize: true
      hidden_units: 256
      num_layers: 2
    reward_signals:
      extrinsic:
        gamma: 0.99
        strength: 1.0
    max_steps: 3000000
    summary_freq: 500
    keep_checkpoints: 5
engine_settings:
  time_scale: 1
  target_frame_rate: -1
  capture_frame_rate: 60
  no_graphics: false
```

Figure 1: Reinforcement learning configuration parameters for the ultrasound probe agent, including network architecture, training settings, and reward signal weights used in Unity ML-Agents.

Script and Reward Definitions:

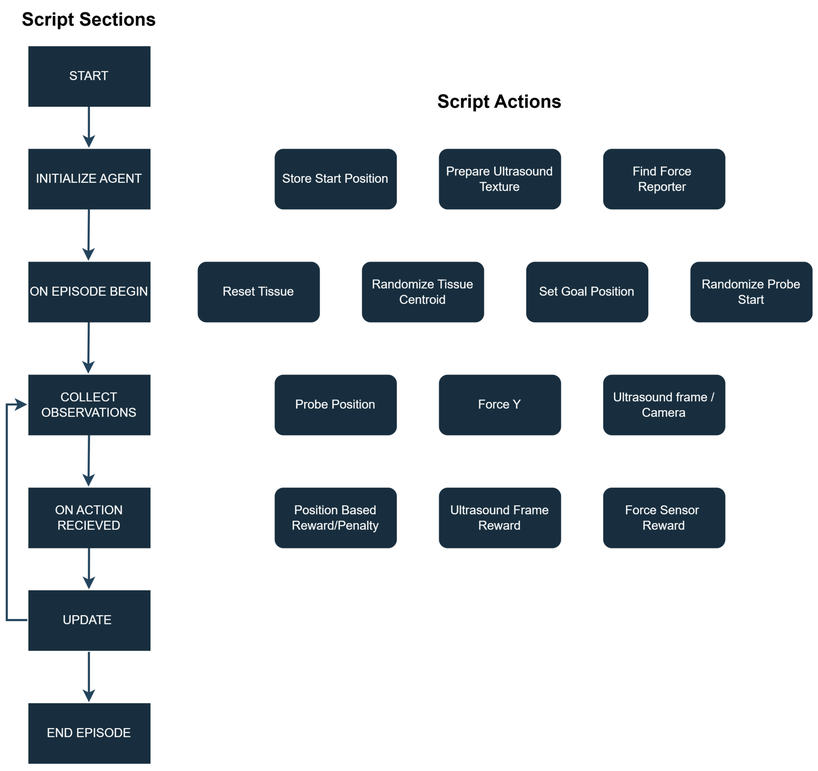

Figure 2: High-level pseudocode outlining the reinforcement learning control loop for the ultrasound probe.

Unity Environment

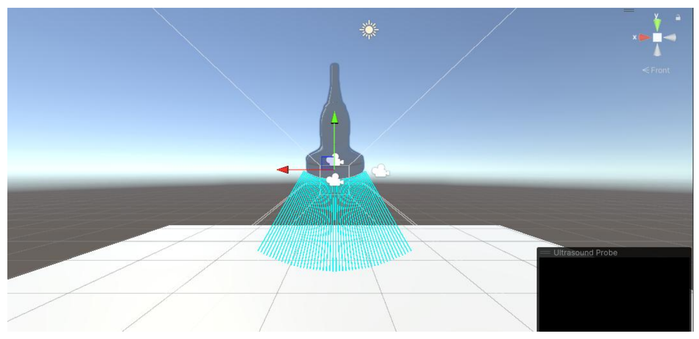

Figure 3: Unity scene view illustrating the simulated robotic ultrasound environment.

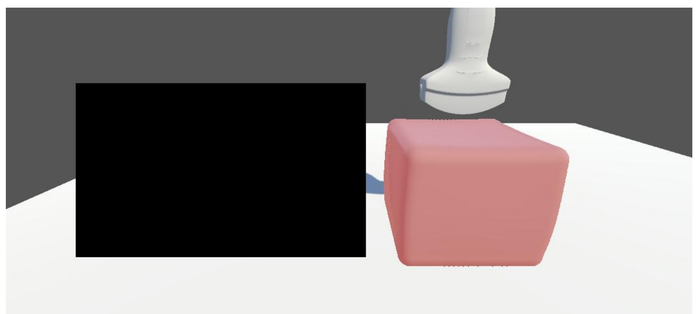

Figure 4: Simulated ultrasound image captured when the probe is not in contact with the tissue.

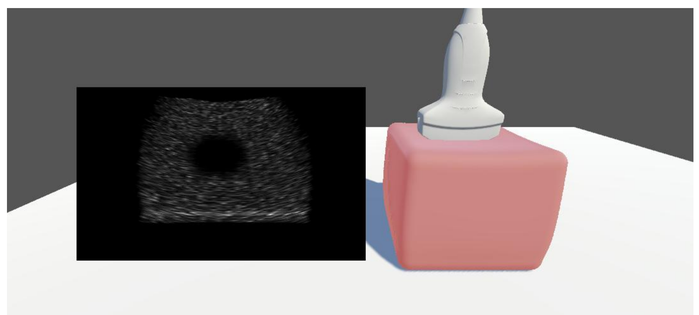

Figure 5: Simulated ultrasound image obtained with proper probe–tissue contact.

Results

```
Mean Reward: 27.620. Std of Reward: 22.839.
Mean Reward: 37.491. Std of Reward: 38.904.
Mean Reward: 31.166. Std of Reward: 36.970.
Mean Reward: 39.647. Std of Reward: 32.792.
 Mean Reward: 24.194. Std of Reward: 30.524.
```

Figure 6: Summary statistics of total episodic rewards reported during training after incorporating position-, image-, and force-based reward components.

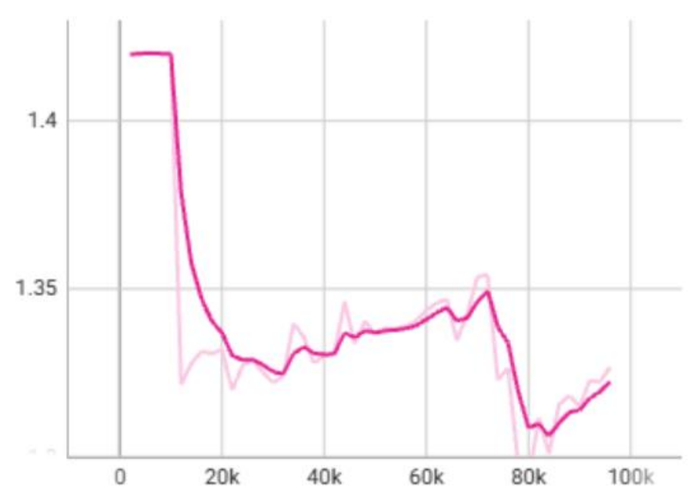

Figure 7: Policy entropy over training steps, showing a gradual reduction in exploration as the agent converges after incorporating position-, image-, and force-based rewards.

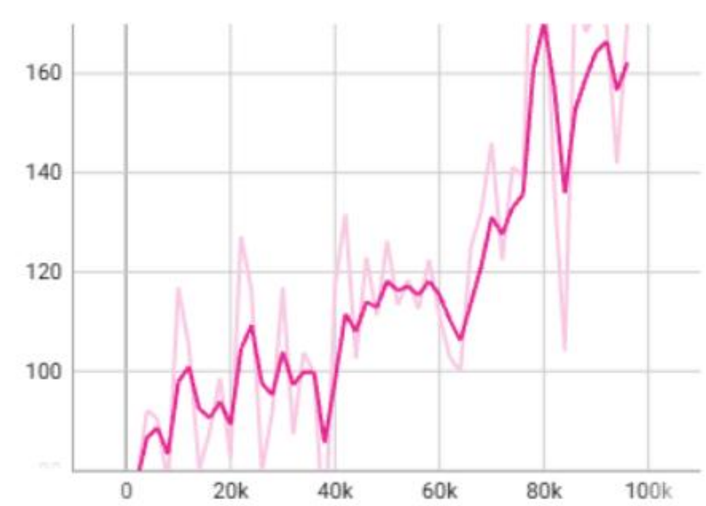

Figure 8: Mean episodic reward as a function of training steps, indicating improved scanning performance and policy learning when position-, image-, and force-based rewards are combined.

Discussion

How do position-based rewards affect probe performance?

- Guides the probe toward target positions in simulation.
- Simple and straightforward to implement and interpret.
- Alone, does not account for tissue deformation or image appearance.
- Limited practical utility for real-world ultrasound scanning without additional rewards.

How do image-based rewards affect probe performance?

- Guides the probe toward the target using ultrasound image cues instead of just positions.
- More practically accurate for real-world scanning tasks.
- Encourages the probe to maintain sufficient contact with the tissue for quality imaging.

How do force-based rewards affect probe performance?

- Prevents the probe from pressing too deeply, protecting tissue and maintaining safety.
- Encourages the probe to maintain consistent contact with the tissue.
- Works together with image- and position-based rewards to improve overall scanning performance.

Conclusions

- Position-based rewards are the most effective during training, guiding the agent with exponential distance shaping.
- Desired image and force signals naturally emerge once the probe reaches the correct region.
- After training, the agent can rely on image and force cues alone, which is essential when the true target position is unknown.

Future Work

- Perform zero-shot testing on ultrasound probe using an actual robot.
- Explore scaling to more complex ultrasound scanning tasks and diverse tissue types.

References

[1] Y. Ou and M. Tavakoli, "Learning Autonomous Surgical Irrigation and Suction with the da Vinci Research Kit Using Reinforcement Learning," arXiv (Cornell University), Nov. 2024, doi: https://doi.org/10.48550/arxiv.2411.14622.

[2] Y. Ou and M. Tavakoli, "CRESSim-MPM: A Material Point Method Library for Surgical Soft Body Simulation with Cutting and Suturing," arXiv (Cornell University), Feb. 2025, doi: https://doi.org/10.48550/arxiv.2502.18437.

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